Joint Detection, Channel Estimation and Interference Nulling for Terrestrial-Satellite Downlink Co-Existence in the Upper Mid-Band

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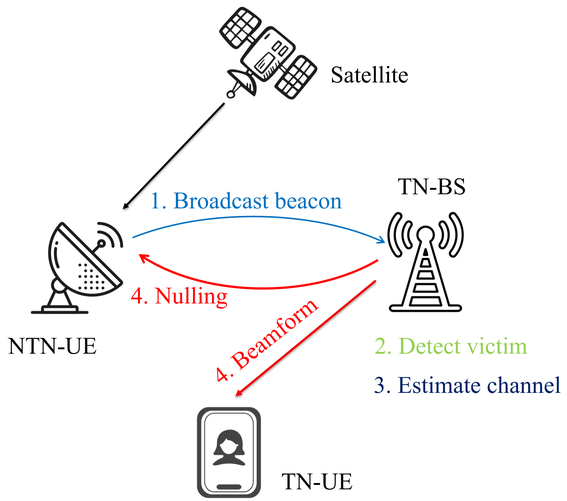
Motivation

Satellites: Growing Fast…and Using FR3

- **Problem:** Interference from TN-gNB to NTN-UE
- NTN and TN systems may be different:
  - Operators: E.g., T-Mobile and Starlink
  - Different standard: 3GPP cellular Vs. Proprietary
- **Challenge:** NTN UE locations unknown a priori:
  - Must be estimated

What is the effect of estimation error on nulling?

Proposed interference nulling framework

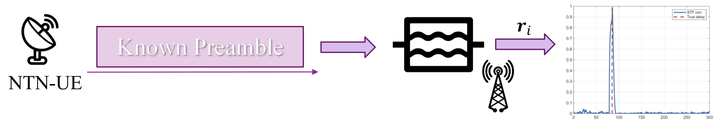


- NTN-UEs broadcast beacon or preamble signals.
- TN-gNB detect the victim NTN-UEs.
- TN-gNB estimate the downlink channel.
- TN-gNB beamform to the desired users with nulls

References:

1. S. Kang, M. Mezzavilla, S. Rangan, A. Madanayake, S. B. Venkatakrishnan, G. Hellbourg, M. Ghosh, H. Rahmani, and A. Dhananjay, "Cellular wireless networks in the upper mid-band," IEEE Open Journal of the Communications Society, 2024
2. J. Hoydis, S. Cammerer, F. Ait Aoudia, M. Nimier-David, L. Maggi, G. Marcus, A. Vem, and A. Keller, "Sionna," 2022, https://nvlabs.github.io/sionna/.
3. S. Kang, G. Geraci, M. Mezzavilla, and S. Rangan, "Terrestrial-satellite spectrum sharing in the upper mid-band with interference nulling," in ICC 2024 - IEEE International Conference on Communications, 2024, pp. 5057–5062.
4. 3GPP, "Solutions for NR to Support Non-Terrestrial Networks (NTN)," 3rd Generation Partnership Project (3GPP), Tech. Rep. TR 38.821 V16.2.0, Mar. 2023, release 16.

Detection, Estima tion and Nulling



- TN-gNB declare a victim exists if the peak energy exceeds a threshold accroding to false alarm goal:

$$\|\boldsymbol{z}_i(\tau)\|^2 \geq t\boldsymbol{N_0}$$

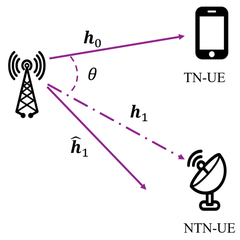
- Estimates the channel by equalizing the preamble:

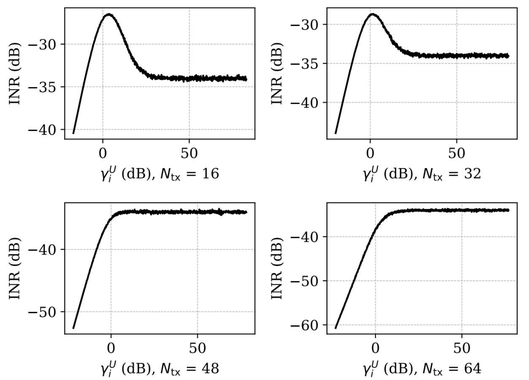
$$\widehat{\boldsymbol{h}}_i = \frac{a_i^*}{|a_i|^2}\boldsymbol{r}_i = \boldsymbol{h}_i + \boldsymbol{v}_i,\ \boldsymbol{v}_i \sim \mathcal{CN}\left(0, \frac{N_0}{\mathcal{E}_u}\boldsymbol{I}\right)$$

- Beamformer design:

$$\widehat{\boldsymbol{w}} = \arg\max_{\|\boldsymbol{w}\|=1} \left[\left|\boldsymbol{w}^*\widetilde{\boldsymbol{h}}_0\right|^2 - \lambda\sum_{i=1}^{K}\left|\boldsymbol{w}^*\widehat{\boldsymbol{h}}_i\right|^2\right]$$

Analytical Insights – Single Victim





- Spatial correlation

$$\rho = \frac{\boldsymbol{h}_0^*\boldsymbol{h}_1}{\|\boldsymbol{h}_0\|\|\boldsymbol{h}_1\|}$$

- Beamforming with $\widehat{\boldsymbol{h}}_i$

$$\widehat{\boldsymbol{w}} = \frac{1}{\sqrt{1-|\hat{\rho}|^2}}\left[\frac{\boldsymbol{h}_0}{\|\boldsymbol{h}_0\|} - \hat{\rho}\frac{\widehat{\boldsymbol{h}}_1}{\|\widehat{\boldsymbol{h}}_1\|}\right]$$

- Downlink symbol energy: $\mathcal{E}_D$
- Uplink sequence energy: $\mathcal{E}_U$

Victim is far — Weak Intefer — Poor estimation

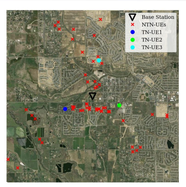
$$\mathbb{E}[\boldsymbol{INR}] \approx \frac{\mathcal{E}_D}{\mathcal{E}_U}\gamma_1^U|\boldsymbol{\rho}|^2\boldsymbol{N_t}$$

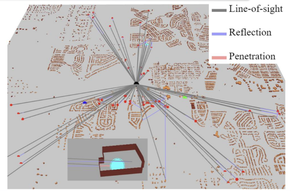
Victim is close — Strong interfer — Good estimation

$$\mathbb{E}[\boldsymbol{INR}] \approx \frac{\mathcal{E}_D}{\mathcal{E}_U}$$

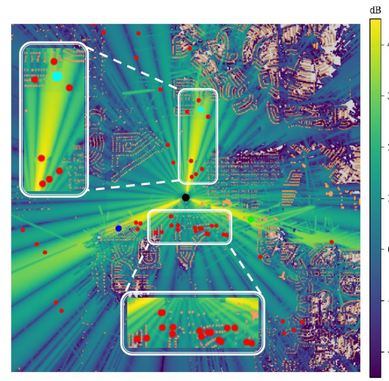
Simulation Settings

Radio Parameters	Values
Fc and B	10 GHz and 100 MHz
Preamble Duration	20 μs
P_fa	10^(-8)
Base Station (BS)	4 × 16 URA, 3 sectors
Handheld (TN-UE)	(1, 1, 2) Omni-directional
VSAT (NTN-UE)	60 cm diameter, Directional

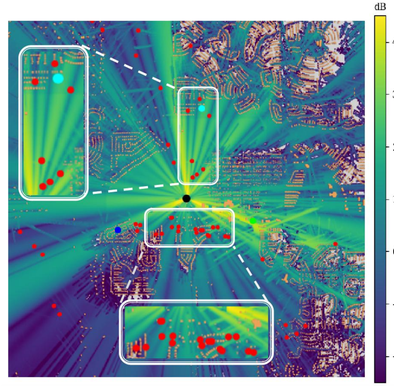




Beamforming Effect

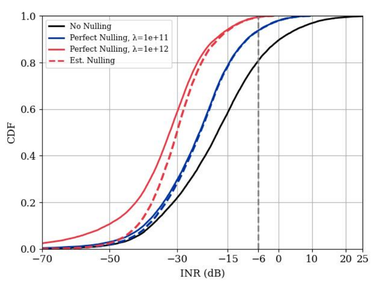


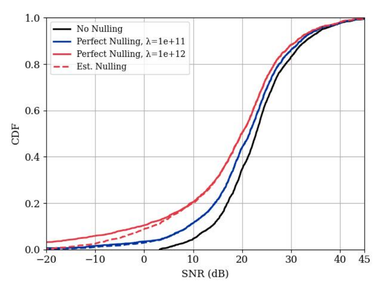
❑SNR Radio map before Nulling



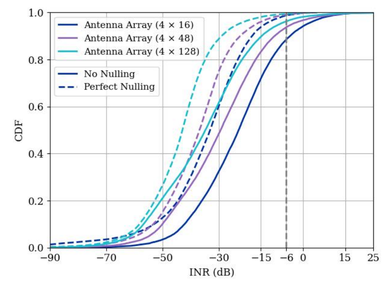
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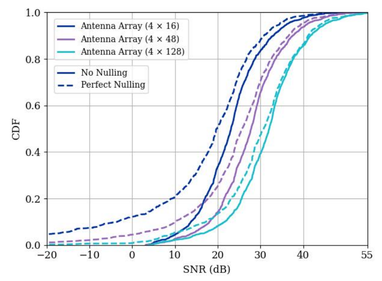
Results – INR Reduction and Massive MIMO Necessity





- No nulling: ~20 % of NTN-UEs have INR > –6 dB.
- Higher λ→ stronger interference suppression but larger SNR penalty
- Trade-off: interference mitigation vs. desired signal quality.





- λ = 10^11, Increasing array size from 4×16 to 4×128
- **INR:** Stronger suppression (dashed curves shift left in INR CDF).
- **SNR:** Retains desired signal quality (curves stay close to no-nulling baseline).